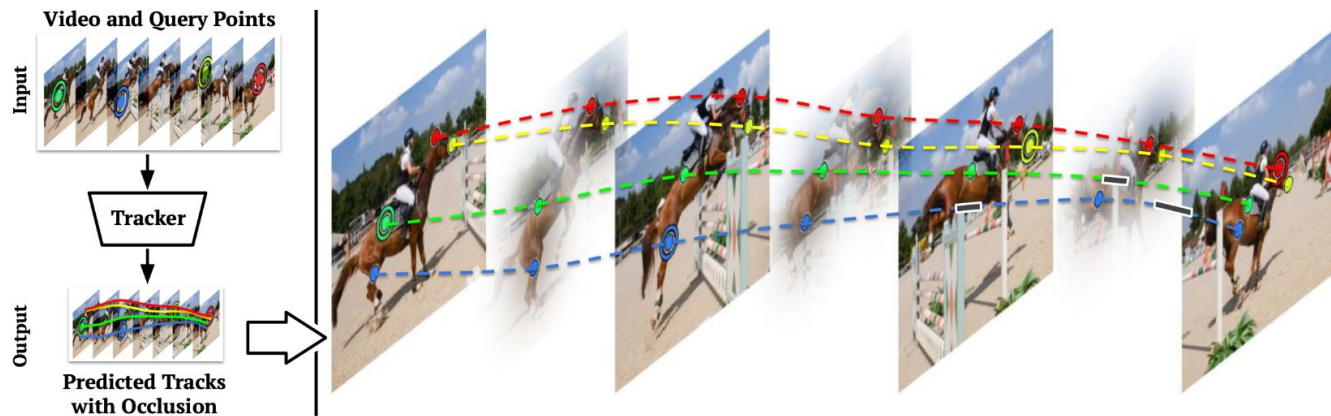


Point Tracking in Egocentric Videos Using Foundation Models

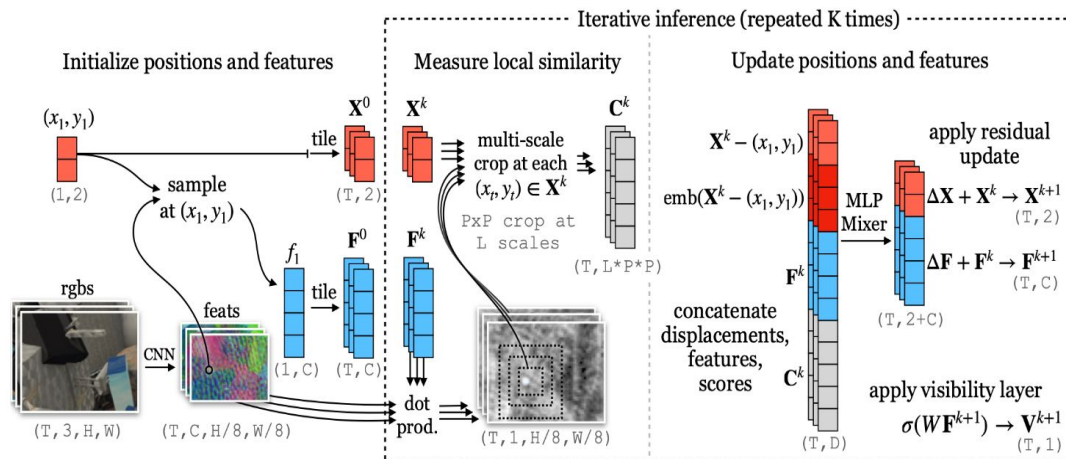
Denis Mbey Akola
Sidhartha Reddy Potu

Introduction



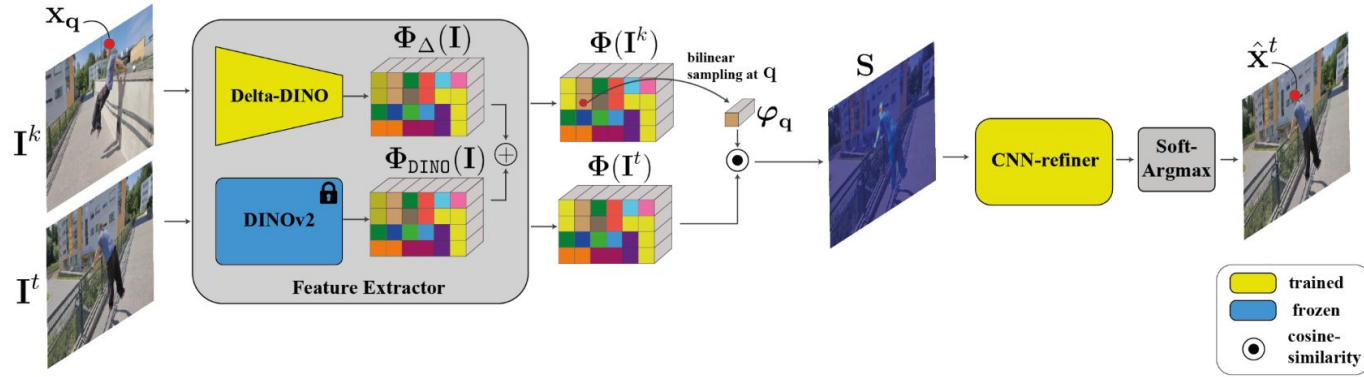
Point tracking involves detecting and following distinct points across video frames to analyze motion and understand scene dynamics.

Related Works - PIPS



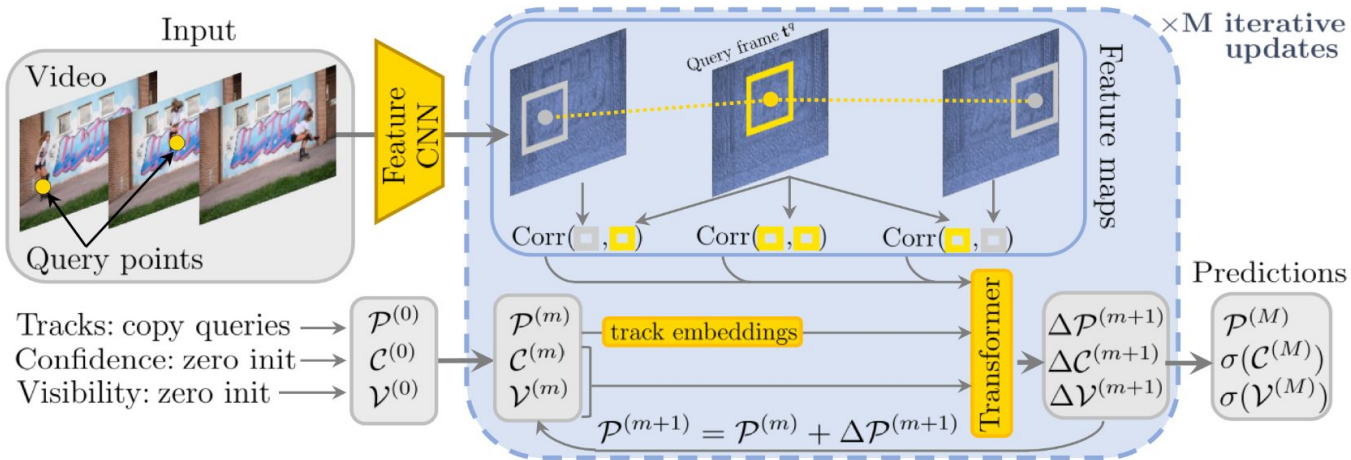
- PIPs fuse cost volumes with deep temporal reasoning for multi-frame tracking.
- Survives long occlusions using iterative inference and trajectory-aware priors.

Related Works - DinoTracker



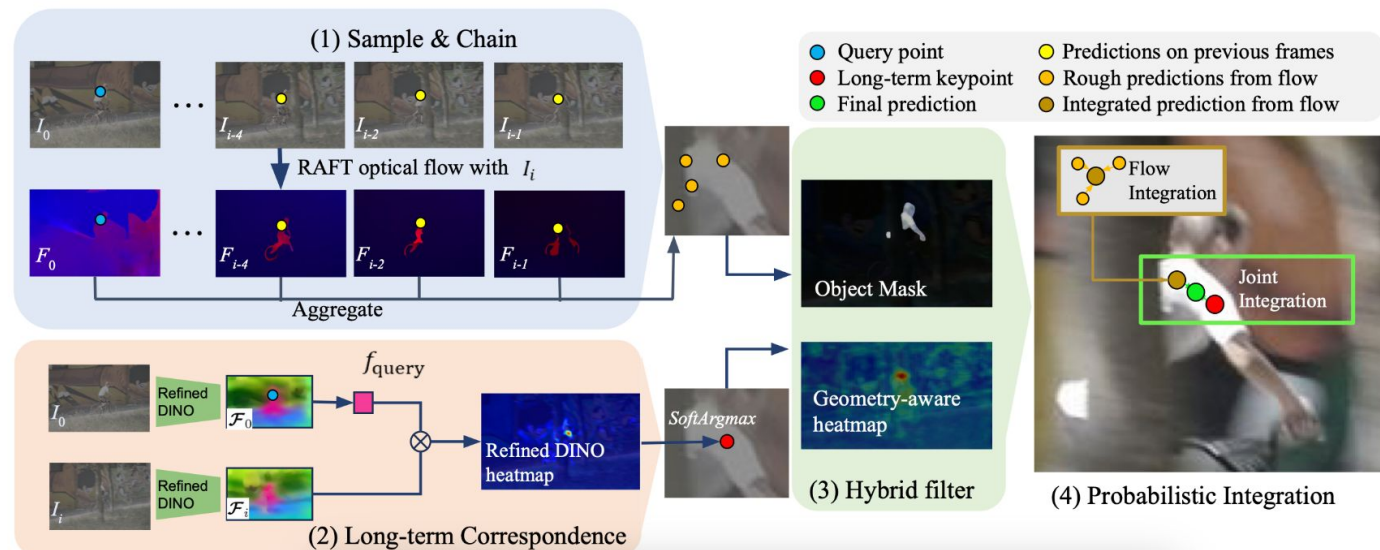
Delta-DINO refines frozen DINOv2 features to improve temporal consistency and localization.

Related Work - CoTracker3



- Jointly tracks multiple points and predicts their visibility and confidence using a space-time transformer.
- Handles occlusions and recovers long-range trajectories with end-to-end optimization.

Related Works -ProTracker



- Tracks dense keypoints robustly through occlusions by leveraging long-term correspondence from Dino Tracker and Optical Flow.

Egocentric Videos

First-person videos captured from wearable cameras (e.g., glasses, head-mounted GoPros)

Challenges of Applying Third-Person Deep Learning Methods to Egocentric Videos

- Viewpoint Bias
- Occlusions and Hand Presence
- Limited Scene Context
- Temporal Irregularities
- Appearance Shift
- Bias in Pre-Training

Datasets : Epic-kitchens, Ego4D, EgoPoints

EgoPoints: Advancing Point Tracking for Egocentric Videos Ahmad Darkhalil, Rhodri Guerrier, Adam W. Harley, Dima Damen (WACV 2025)

Point tracking in Egocentric videos

Given a sequence of video frames $\{I_1, I_2, \dots, I_N\}$, and a set of keypoints $\{p_1^t, p_2^t, \dots, p_N^t\}$ in frame t , point tracking aims to estimate correspondences:

$$p_i^{t+1} = p_i^t + \Delta p_i^t \quad \text{for } i = 1, \dots, N$$

where Δp_i^t is the displacement of keypoint i from frame t to $t+1$.

Evaluation of Third-Person Trackers

Model	TAP-Vid-DAVIS		EgoPoints		
	$\delta_{\text{avg}} \uparrow$	$\delta_{\text{avg}} \uparrow$	ReID $\delta_{\text{avg}} \uparrow$	OOVA $\uparrow$	IVA $\uparrow$
PIPs++	64.0	36.9	14.6	50.4	89.2
CoTracker	74.7	38.5	4.8	81.4	73.4
BootsTAPIR Online	65.2	39.6	0.0	0.0	100.0
LocoTrack	75.3	59.4	0.1	0.2	99.9
CoTracker v3	77.2	50.0	15.0	31.8	99.3

$\delta^{(\text{avg})}$: Avg. accuracy over pixel error thresholds

ReID $\delta^{(\text{avg})}$: Avg. re-ID accuracy over error thresholds

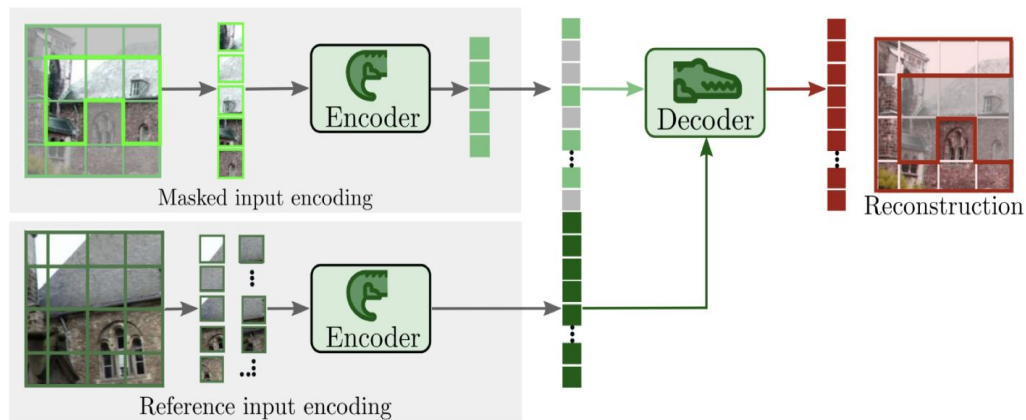
OOVA: Accuracy when target is out of view

IVA: Accuracy when target is in view

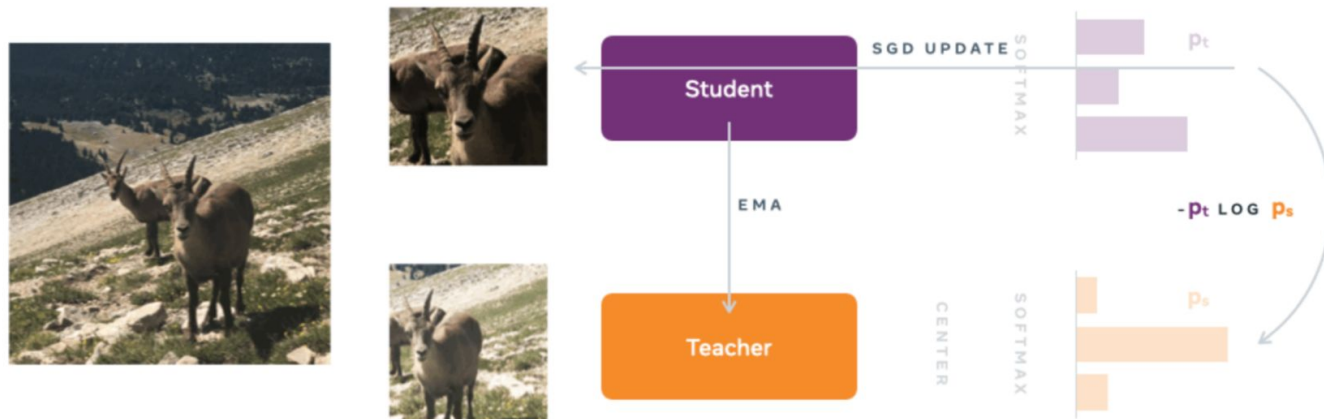
Spoiler Alert 🚨: Third-person trackers struggle with egocentric tracking due to occlusions and motion challenges.

Foundation Models - CroCov2

CroCo: a cross-view completion pre-training method



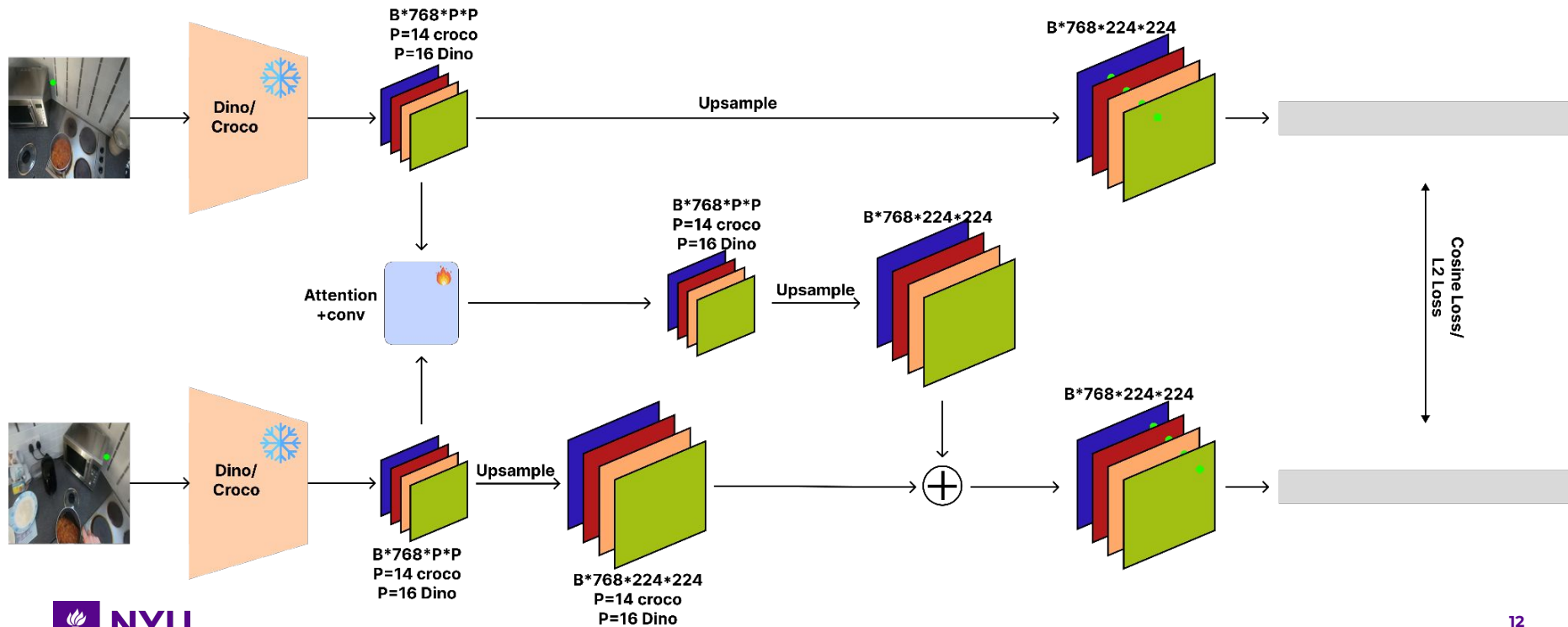
Foundation Models - DINOv2



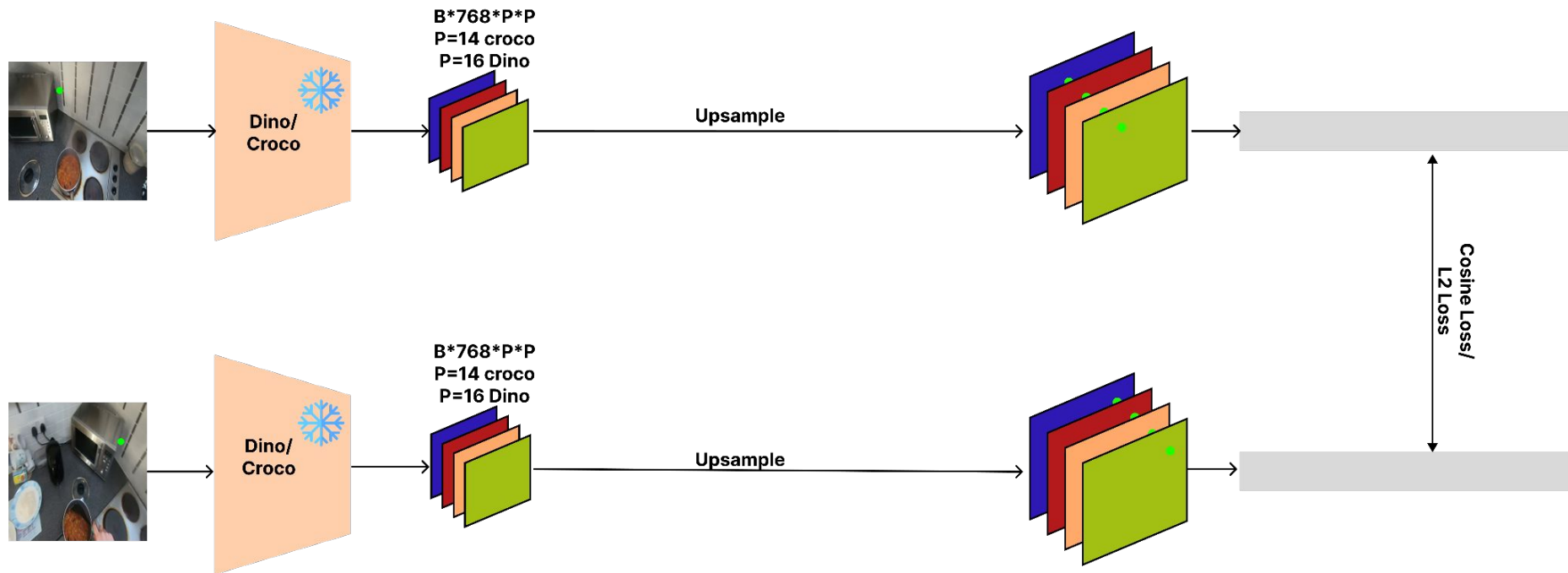
DINOv2 uses a self-supervised objective to learn view-invariant features via a teacher-student setup without labels.

DINO: Emerging Properties in Self-Supervised Vision Transformers (Caron et al., 2021)

Method



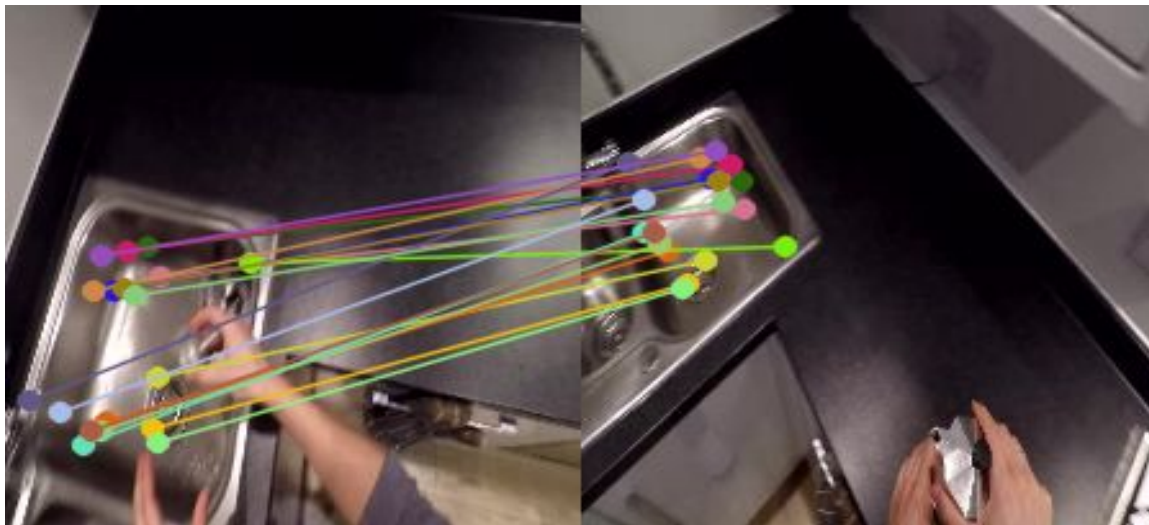
Method



Dataset Curation - EPIC Fields

Clustering

Cluster camera centers and retrieve diverse frame pairs. Estimate overlap using COLMAP mesh visibility (Sparse key points)



Dataset Curation - EgoPoints

We select frame pairs from EgoPoints videos with annotated point tracks using a windowing approach.



Model Training Objective

Let $\mathbf{f}_1, \mathbf{f}_2 \in \mathbb{R}^d$ be the feature vectors at corresponding keypoints.

$$\mathcal{L}_{L2} = \|\mathbf{f}_1 - \mathbf{f}_2\|_2^2$$

$$\mathcal{L}_{\cos} = 1 - \frac{\mathbf{f}_1 \cdot \mathbf{f}_2}{\|\mathbf{f}_1\|_2 \|\mathbf{f}_2\|_2}$$

Experiments and Results

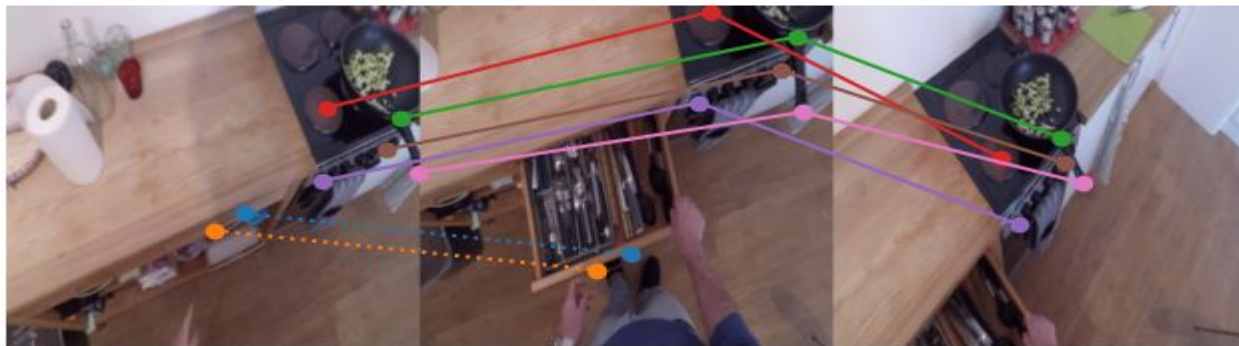
Evaluation on a subset of EgoPoints Test Set

Model	Loss	Delta	acc@8px	acc@16px	acc@24px	avg
Croco	L2	✓	12.770	24.09	32.33	23.06
Croco	L2	✗	13.890	25.28	34.13	24.44
Croco	Cosine	✗	20.960	35.66	43.54	33.39
Croco	SSL loss	✗	12.460	24.35	31.78	22.87
Croco (Pre-trained)	—	✗	20.100	34.31	42.13	32.18
Dinov2	Cosine	✗	4.15	12.17	17.69	11.34
Dinov2	L2	✓	1.580	6.24	10.41	6.08
Dinov2	L2	✗	3.780	10.74	15.12	9.09
Dinov2 (Pre-trained)	—	✗	19.570	37.04	47.00	34.54

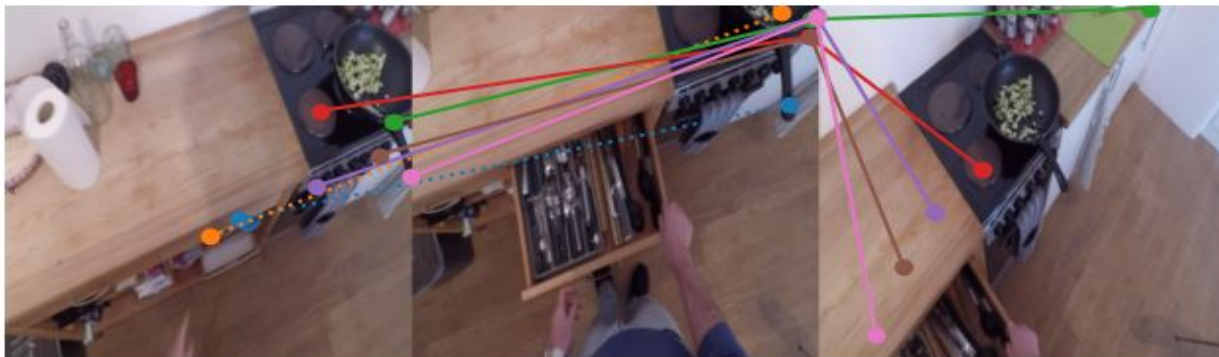
acc@Threshold: Accuracy within a pixel error threshold. **Red** : best, **Blue** : second best

Qualitative results

Ground truth



Prediction from
Dinov2



Future Work

- **Modeling Motion**

Integrate optical flow prediction as a supervisory signal to improve point tracking.

- **Leveraging Pretrained Models**

Systematically evaluate how to best utilize pretrained models for egocentric tracking.

- **Hybrid Approaches**

Combine CroCo and DINOv2 to capture complementary aspects of scene motion.

- **3D Foundation Models**

Explore the use of Dust3r and MAST3R for egocentric tracking via 3D reasoning, multi-scale features, and task-aligned training objectives.

Conclusion & Takeaways

- **Egocentric Tracking is Challenging**
Involves motion reasoning, matching correspondences, and handling rapid viewpoint changes.
- **Foundation Models Underperform**
Pretrained models struggle out-of-the-box on egocentric point tracking.
- **Mismatch in Objectives**
General training goals don't align well with fine-grained tracking needs.
- **Fine-tuning is Non-trivial**
It's unclear what to adapt, how much, or how to inject the right inductive biases.

Thank you for listening!

Any questions?